

EviSpan: Evidence-Grounded Generative Evaluation for Machine Translation

Anonymous ACL submission

Abstract

Machine translation evaluation becomes more useful when quality scores are accompanied by inspectable evidence. However, many strong MT metrics still return only scalar scores, while recent generative evaluators often provide free-form rationales that are not explicitly tied to verifiable error spans. We introduce EviSpan, a span-grounded generative evaluator that (1) marks erroneous spans in a translation with inline tags, (2) predicts MQM-style error labels, (3) generates a concise rationale, and (4) outputs a segment-level quality score. To train the model without gold rationales, we suggest verifiable rewards derived from MQM annotations, combining format validity, multi-annotator span agreement, and score calibration in a two-stage GRPO curriculum. Experiments on the WMT22 and WMT24 MQM benchmarks show that EviSpan is competitive with a same-size Remedy-R generative baseline on translation scoring while additionally producing explicit span-level evidence. EviSpan achieves competitive weighted character-level span F1 on the WMT22 News-domain MQM subset and is more often preferred over Remedy-R in an LLM-as-a-judge assessment of evaluation feedback. These results indicate that explicit span grounding can make generative MT evaluation more inspectable while remaining competitive on scoring.¹

1 Introduction

Automatic machine translation (MT) evaluation is central to comparing systems, monitoring translation quality, and guiding model development. Segment-level quality scores are especially useful because they provide a compact signal for ranking systems and diagnosing model changes at scale. However, a quality score alone rarely explains

which parts of a translation justify the judgment, making it difficult for users to inspect errors, verify metric decisions or obtain actionable feedback. Human evaluation protocols such as Multidimensional Quality Metrics (MQM; Lommel et al., 2014; Freitag et al., 2021) provide this kind of evidence by marking erroneous spans, assigning error categories, and indicating severity levels. However, MQM annotation requires expert bilingual annotators and is expensive to scale. As a result, many strong automatic MT metrics are still trained and used primarily as scalar scorers, including strong learned neural metrics such as COMET-22 (Rei et al., 2022) and MetricX (Juraska et al., 2023, 2024).

Recent work has sought to address this evidence gap from several directions. Span-aware metrics such as xCOMET (Guerreiro et al., 2024) provide fine-grained error evidence, and correction-oriented systems such as xTower (Treviso et al., 2024) can generate explanations and corrections from externally supplied marked spans. Generative evaluators such as Remedy-R (Tan et al., 2025), in contrast, produce natural-language analyses before assigning scores. However, existing approaches target complementary outputs: xCOMET predicts spans and scores, xTower explains externally supplied marked spans, and Remedy-R generates free-form reasoning before scoring. This motivates the question of how to train a unified generative evaluator that produces local evidence, rationales, and scores together, while using MQM annotations to verify the generated evidence and score.

We address this gap with span-grounded MT evaluation, an evidence-first generative framework in which human-marked error spans serve as verifiable training signals. The model marks predicted erroneous spans with inline tags before generating MQM-style labels, a concise rationale, and a segment-level quality score, making the

¹Code and model checkpoint: Repo hidden for review

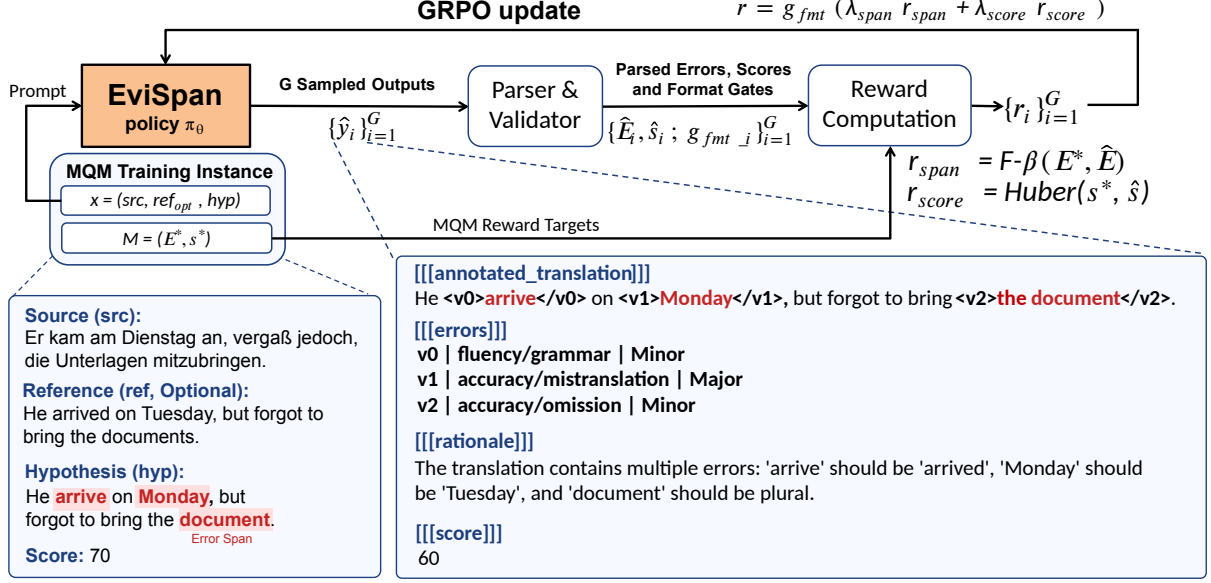


Figure 1: Training pipeline for span-grounded MT evaluation with GRPO. Each MQM training instance provides a prompt input $x = (src, ref_{opt}, hyp)$ and MQM reward targets $M = (E^*, s^*)$. The evaluator receives only x and samples G structured outputs containing tagged error spans, error labels with severities, a rationale, and a final score. A parser and validator extract $\hat{E}_i$, $\hat{s}_i$, and $g_{fmt,i}$ from each output. Reward computation compares the parsed predictions with M using weighted character-level F_β for span agreement and a Huber-style reward for score calibration, gated by $g_{fmt,i}$. The resulting rewards $\{r_i\}_{i=1}^G$ are used for the GRPO update.

local evidence behind the judgment explicit and machine-readable. Without gold rationales, we train the evaluator using verifiable MQM-derived rewards for format validity, multi-annotator span agreement, and score calibration, optimized with a two-stage GRPO (Shao et al., 2024) curriculum. Experiments on WMT22 and WMT24 MQM benchmarks show that our approach remains competitive with the same-size Remedy-R generative baseline in translation scoring while producing explicit span-level evidence with competitive span detection accuracy.

Our contributions are threefold. First, we introduce an evidence-first generative MT evaluator that exposes MQM-style error spans before producing rationales and scores. Second, we design verifiable multi-annotator rewards for span grounding and score calibration, optimized with a two-stage GRPO curriculum without gold rationale supervision. Third, we show on WMT MQM benchmarks that span-grounded generation preserves competitive scoring performance, produces competitive span-level evidence, and yields feedback preferred over Remedy-R in a pairwise LLM-as-a-judge assessment. An evidence-prefix intervention further shows that exposed MQM evidence is actionable for scoring: clamping gold evidence improves segment-level accuracy and

moves predicted scores closer to human MQM scores.

2 Related Work

2.1 Scalar and Preference-based MT Metrics

Learned MT metrics such as COMET-22 (Rei et al., 2022) and MetricX (Juraska et al., 2023, 2024) achieve strong agreement with human judgments by predicting segment-level quality scores from the source, hypothesis, and optionally a reference. Preference-based metrics such as ReMedy (Tan and Monz, 2025) further train reward models from human preferences and show strong performance on WMT MQM benchmarks. These approaches are effective for ranking and scoring translations, but their primary output remains a scalar score, offering limited evidence about which parts of the translation justify the judgment.

2.2 Span-aware MT Evaluation

Several metrics improve interpretability by predicting fine-grained error information. xCOMET (Guerreiro et al., 2024) augments COMET-style sentence-level quality scoring with word-level severity prediction, while AutoMQM (Fernandes et al., 2023) uses LLMs to produce MQM-style annotations. Tagged Span Annotation (Yeom

et al., 2025) represents errors by inserting inline tags around erroneous spans; we adopt this representation as the evidence layer of our generative evaluator. xTower (Treviso et al., 2024) generates explanations and corrected translations from marked error spans, but assumes those spans as external input and relies on GPT-4-distilled supervision. In contrast, EviSpan jointly generates spans, rationales, and scores, and directly optimizes the generated spans and severities with MQM-derived verifiable rewards.

2.3 Generative Reasoning Metrics

LLMs have also been used as generative MT evaluators. GEMBA (Kocmi and Federmann, 2023) showed that large proprietary LLMs can serve as strong prompt-based MT metrics without task-specific training. Remedy-R (Tan et al., 2025) is the most direct comparison to our work. It trains an open LLM to generate reasoning before predicting a final score, without proprietary model distillation or explicit span annotations. In contrast, our work studies a complementary span-supervised setting, where available MQM spans are used to make the model expose structured local evidence before generating the rationale and score.

3 Background

3.1 MQM-based MT Evaluation

Multidimensional Quality Metrics (MQM; Lommel et al., 2014; Freitag et al., 2021) provides fine-grained human annotations for MT evaluation by marking erroneous spans, assigning error categories, and labeling their severities. Since segment-level quality is derived from these local annotations, MQM spans serve as verifiable evidence for the final judgment rather than merely post-hoc explanations. We therefore use MQM-derived scores to calibrate predicted quality scores and use human-marked spans and severities to define verifiable rewards for span-grounded generation.

3.2 WMT Meta-evaluation and Span Metrics

MT metrics are commonly evaluated by their agreement with human judgments at both the system and segment levels. For system-level evaluation, we follow the official WMT protocol used for each benchmark: pairwise accuracy (Acc) for WMT22 MQM and Soft Pairwise Accuracy (SPA; Thompson et al., 2024)

for WMT24 MQM. For segment-level evaluation, we report tie-calibrated accuracy, (acc_{eq}^* ; Deutsch et al., 2023). We additionally evaluate weighted character-level F1 (Zerva et al., 2024) for predicted error spans, measuring overlap with human MQM spans while accounting for severity matches and partial severity disagreements.

3.3 Verifiable Rewards and GRPO

Our task lacks gold rationales, but several desired properties of the output can be automatically verified, including format validity, span agreement, and score calibration. Following reinforcement learning with verifiable rewards (RLVR; Wen et al., 2025), we define deterministic rewards from MQM annotations instead of training a learned reward model. We optimize these rewards with Group Relative Policy Optimization (GRPO; Shao et al., 2024), which compares multiple sampled outputs for the same input and updates the policy based on their relative quality without a separate value model.

4 EviSpan

4.1 Span-grounded MT Evaluation

We formulate MT evaluation as evidence-first conditional generation. Let *inst* denote a fixed task instruction that specifies the evaluation criteria, input fields, and required output sections. Given a source sentence *src*, an optional reference ref_{opt} , and a translation hypothesis *hyp* for a single segment, the model receives $x = \langle inst, src, ref_{opt}, hyp \rangle$ and generates a structured evaluation *y* in which local error evidence precedes the rationale and final score. The output consists of four sections: `[[[annotated_translation]]]`, `[[[errors]]]`, `[[[rationale]]]`, and `[[[score]]]`. The annotated translation copies *hyp* while inserting paired inline tags (e.g., `<v0>...</v0>`) around predicted erroneous spans. The error section maps each tag to a coarse MQM-style error type and a severity label, MINOR or MAJOR. The rationale briefly explains the tagged evidence, and the score section outputs a segment-level quality score $\hat{s} \in [0, 100]$ for *hyp*. We collectively refer to the non-score components—the tagged annotated translation, structured error list, and rationale—as evaluation feedback.

A deterministic parser removes tags to verify

that the annotated translation preserves the original hypothesis, then extracts the predicted span set $\hat{E}$ and score $\hat{s}$. Error types are generated for readability, while training rewards evaluate span localization, severity, and score calibration. Rationales are not directly supervised; the evidence-first format encourages them to refer to preceding tagged spans and error labels.

4.2 Multi-annotator Verifiable Rewards

For each input x , let

$$\mathcal{M}(x) = \{(E_j^*, s_j^*)\}_{j=1}^{J_x}$$

be the retained MQM annotations, where E_j^* is annotator j 's error-span set and s_j^* is the corresponding segment-level quality score. We keep all retained annotators rather than collapsing them into a single gold span set, and define

$$E^*(x) = \{E_j^*\}_{j=1}^{J_x}, \quad \bar{s}^*(x) = \frac{1}{J_x} \sum_{j=1}^{J_x} s_j^*.$$

Given a parsed model output $(\hat{E}, \hat{s})$, we compute a verifiable reward

$$r(y; x) = g_{\text{fmt}}(y, x) r_{\text{task}}(y; x),$$

with

$$r_{\text{task}}(y; x) = \lambda_{\text{span}} r_{\text{span}}(\hat{E}, E^*(x)) + \lambda_{\text{score}} r_{\text{score}}(\hat{s}, \bar{s}^*(x)).$$

The binary gate g_{fmt} is 1 only when the output is parseable, contains all required sections, preserves the original hypothesis after tag removal, uses valid paired tags, and produces a score in $[0, 100]$. The span reward averages weighted character-level overlap with each retained annotator:

$$r_{\text{span}}(\hat{E}, E^*(x)) = \frac{1}{J_x} \sum_{j=1}^{J_x} F_{\beta}^{\text{char}}(\hat{E}, E_j^*).$$

Overlapping characters receive full credit when severity labels match and half credit otherwise. During training, we use $F_{0.5}$ to emphasize precision and discourage unsupported over-tagging.

The score reward calibrates $\hat{s}$ to the annotator-averaged score. With $d = \min(|\hat{s} - \bar{s}^*(x)|, D)$, we define a Huber-style penalty

$$\rho_{\delta}(d) = \begin{cases} \frac{1}{2}(d/\delta)^2, & d \leq \delta, \\ d/\delta - \frac{1}{2}, & d > \delta, \end{cases}$$

and normalize it as

$$r_{\text{score}}(\hat{s}, \bar{s}^*(x)) = 1 - \frac{\rho_{\delta}(d)}{\rho_{\delta}(D)}.$$

Thus $r_{\text{score}} = 1$ for an exact score match and approaches 0 when the clipped error reaches D .

4.3 GRPO Optimization

We optimize the evaluator with GRPO using the verifiable reward above. For each input x , the rollout policy samples G structured evaluations $\{y_i\}_{i=1}^G$; each output is parsed, checked by g_{fmt} , and assigned reward $r(y_i; x)$. We compute group-relative advantages

$$A_i = \frac{r(y_i; x) - \mu_r}{\sigma_r + \epsilon},$$

where μ_r and σ_r are the mean and standard deviation of rewards within the group. The policy is then updated with the standard clipped GRPO objective and a KL penalty to the frozen reference policy.

To reduce interference between evidence generation and score calibration, we use a two-stage curriculum. Stage 1 uses the format gate and span reward only, encouraging the model to produce valid tagged spans before learning final score calibration. Stage 2 initializes from the selected Stage-1 checkpoint and adds the score reward, jointly optimizing span grounding and score calibration.

5 Experiments

We evaluate EviSpan along three axes: translation scoring, error span detection, and the usefulness of evaluation feedback. Models are trained on WMT20 MQM annotations (Freitag et al., 2021), while WMT22 (Freitag et al., 2022) and WMT24 (Freitag et al., 2024) MQM test sets are used for translation-scoring meta-evaluation, and the WMT22 News-domain MQM subset is used for span-level evaluation.

5.1 Data, Prompts and Benchmarks

We train on the WMT20 MQM data, following the controlled-data setting of Remedy-R (Tan et al., 2025). The data covers English-German and Chinese-English and provides professional segment-level quality judgments together with fine-grained error-span annotations. For each source segment, multiple MT system outputs are

Type	Method	θ	Avg	Acc	acc_{eq}^*
Scalar Metrics	COMET-22	0.5B	68.7	82.8	54.5
	MetricX-XXL	13B	71.9	85.0	58.8
	ReMedy-9B	9B	75.1	91.2	58.9
	ReMedy-9B-QE	9B	73.6	89.4	57.8
LLM Judges	MQM-APE (Mistral)	8×22B	71.3	88.3	54.2
	GEMBA-DA (GPT-4)	>100B	72.7	89.8	55.6
	Remedy-R-14B	14B	72.4	88.7	56.0
	Remedy-R-32B	32B	73.4	91.6	55.2
Ours	EviSpan-14B	14B	73.6	90.2	57.0
	EviSpan-14B-QE	14B	72.9	90.1	55.8

(a) WMT22 MQM

Type	Method	θ	Avg	SPA	acc_{eq}^*
Scalar Metrics	COMET-22	0.5B	68.8	82.3	55.4
	XCOMET-XXL	10.7B	71.9	86.1	57.6
	XCOMET-XXL-QE	10.7B	69.5	83.3	55.7
	MetaMetrics-MT	ensemble	72.5	85.3	59.6
	MetricX-24-Hybrid	13B	72.1	85.6	58.5
	MetricX-24-Hybrid-QE	13B	71.4	84.9	58.0
	ReMedy-9B	9B	72.9	85.9	60.0
	ReMedy-9B-QE	9B	72.1	84.9	59.3
LLM Judges	GEMBA-ESA (GPT-4)	>100B	71.1	84.6	57.6
	Remedy-R-14B	14B	72.6	87.9	57.2
	Remedy-R-32B	32B	70.6	85.3	56.2
Ours	EviSpan-14B	14B	73.2	88.0	58.5
	EviSpan-14B-QE	14B	72.9	87.6	58.3

(b) WMT24 MQM

Table 1: Main results on WMT22 and WMT24 MQM test sets. Acc denotes system-level pairwise accuracy for WMT22, and SPA denotes system-level Soft Pairwise Accuracy for WMT24. acc_{eq}^* denotes segment-level tie-calibrated pairwise accuracy. Avg is the arithmetic mean of the system-level metric and acc_{eq}^* , and QE denotes reference-free evaluation with same model. Baseline numbers are from published reports unless otherwise noted; EviSpan results are computed by us as means of final meta-evaluation scores over repeated runs. Bold marks the best numerical score within each method family, with Ours grouped with Generative/LLM Judges.

annotated by up to three human raters. To handle rare translation string mismatches among raters for the same source-system item, we select the canonical hypothesis as the translation string shared by the largest number of raters and retain only annotations that match it. After filtering, each training instance contains one to three retained MQM annotations, all of which are used for reward computation.

We construct both reference-based and reference-free prompts. The reference-based prompt includes the source sentence, translation hypothesis, and reference translation, whereas the reference-free prompt omits the reference. MQM error spans, severity labels, and MQM-derived scores are used only to compute rewards and are never shown in the prompt. Full prompt templates are provided in Table 7. We convert MQM penalty scores to 0–100 quality scores as $score = \text{clip}(100 + 4 \cdot \text{MQM}, 0, 100)$.

Final scoring evaluation uses the official WMT22 and WMT24 MQM test sets from `mt-metrics-eval`². WMT22 covers English-German, English-Russian, and Chinese-English; WMT24 covers English-German, English-Spanish, and Japanese-Chinese. For error-span detection, we evaluate on the WMT22 News-domain MQM subset following the xCOMET (Guerreiro et al., 2024) setting. We report both reference-based and reference-free

(Quality Estimation, QE) results.

5.2 Training Setup

We use Qwen2.5-Instruct (Qwen Team, 2024) as the backbone family and take Qwen2.5-14B-Instruct as the main model, enabling direct comparison with the 14B Remedy-R baseline. EviSpan is trained with LoRA (Hu et al., 2022) parameter-efficient fine-tuning to make GRPO training with multiple rollouts feasible for the 14B backbone.

Training follows the two-stage GRPO curriculum described in Section 4.3. We start directly from the instruction-tuned backbone without supervised fine-tuning on serialized MQM outputs. We also tested supervised warm-up on serialized MQM outputs before GRPO, but found that it did not provide the best scoring-grounding trade-off; details are reported in Appendix B.2. Both stages use the same WMT20 MQM training data with different shuffling seeds and are trained for up to one epoch with early stopping. For checkpoint selection, we use a held-out WMT23 MQM validation set consisting of 600 En-De and 600 Zh-En examples sampled from the validation data. Early stopping is based on the validation reward computed with the same reward configuration as the corresponding training run, including the active reward components and their weights. Main hyperparameters are summarized in Table 8.

²<https://github.com/google-research/mt-metrics-eval>

5.3 Evaluation Protocols and Baselines

We parse each generated output using the pre-defined section headers. The final quality score is read from the `[[[score]]]` section, and predicted error spans are extracted from the tagged `[[[annotated_translation]]]` section together with severity labels from `[[[errors]]]`. If the score cannot be parsed, we assign a score of 0. If span fields cannot be parsed, the output is treated as predicting no error spans. Table 9 shows that parsing failures are rare across both test sets. Generated rationales are not used for WMT scoring or span-F1, and are evaluated separately in the feedback-usefulness study.

For translation scoring, we follow the official WMT Metrics Shared Task protocol. On WMT22, we report system-level Pairwise Accuracy (Acc) and segment-level tie-calibrated pairwise accuracy (acc_{eq}^* ; Deutsch et al., 2023). On WMT24, we report system-level Soft Pairwise Accuracy (SPA; Thompson et al., 2024) and segment-level acc_{eq}^* . We use the average of system- and segment-level scores as the main summary score.

For span detection, we report weighted character-level F1 on the WMT22 News domain MQM subset. Predicted inline tags are converted into character-level spans over the original hypothesis and compared with gold MQM spans. Overlapping characters receive full credit when severity labels match and half credit when they differ.

For translation scoring, we compare against representative scalar, preference-based, span-aware, and generative evaluators, including COMET-22 (Rei et al., 2022), MetricX (Juraska et al., 2023, 2024), xCOMET-XXL (Guerreiro et al., 2024), ReMedy (Tan and Monz, 2025), GEMBA (Kocmi and Federmann, 2023), and Remedy-R (Tan et al., 2025). Remedy-R is the most direct generative baseline because it uses the Qwen2.5 model family and generates natural-language analyses followed by final scores, but is trained without explicit error span supervision. For span detection, we compare against xCOMET-XL, xCOMET-XXL, and AutoMQM (Fernandes et al., 2023) with GPT-3.5 (OpenAI, 2022) and GPT-4 (OpenAI, 2023) prompting. For paired statistical significance testing, we use mt-metrics-eval with the Perm-Both test (Deutsch et al., 2021) using 1,000 permutations and block size 100. Significance is assessed separately for Sys and Seg, while Avg is

treated as a descriptive summary.

5.4 LLM-as-a-Judge Assessment of Evaluation Feedback

We evaluate whether the feedback defined in Section 4.1 is useful beyond scores and span overlap with a rubric-based pairwise LLM-as-a-judge protocol (Liu et al., 2023; Zheng et al., 2023). Our primary comparison is against Remedy-R, a free-form generative evaluator that produces natural-language reasoning before assigning a final score. From WMT22 MQM English-German, English-Russian, and Chinese-English, we sample 300 quality-stratified examples per language pair and compare EviSpan feedback with Remedy-R feedback for the same source, hypothesis, and reference. We remove final scalar scores from both outputs so that the judgment focuses on feedback content rather than numeric calibration.

We also conduct a supplementary comparison with xTower (Treviso et al., 2024), a strong correction-oriented feedback baseline, in Appendix D; unlike Remedy-R, xTower receives external xCOMET-XL marked spans and may output correction-oriented feedback, so we treat it as a challenging diagnostic rather than the primary comparison.

The judge compares anonymized candidates across six criteria in randomized A/B order, and each comparison is repeated with the candidate order reversed to reduce position and presentation biases. To reduce presentation, position, and length-related biases (Zheng et al., 2023; Dubois et al., 2024), we instruct the judge not to reward length or formatting alone and evaluate each comparison twice with the candidate order reversed. We report EviSpan-win, Remedy-R-win, and tie percentages in the main results. Details of the Remedy-R judge evaluation, including prompts, schemas, sampling, agreement, and bias diagnostics, are provided in Appendix C. The supplementary xTower comparison is reported in Appendix D.

6 Evaluation Results and Analysis

We evaluate EviSpan along two complementary dimensions: agreement with human MT judgments and the usefulness of its inspectable span-grounded feedback. Results cover WMT scoring, error span detection, feedback usefulness, and reward/curriculum ablations.

METRIC	En-De	En-Ru	Zh-En	Avg.
AutoMQM (GPT-3.5)	0.160	0.166	0.143	0.156
AutoMQM (GPT-4)	0.257	0.281	0.248	0.262
xCOMET-XL	0.290	0.281	0.237	0.269
xCOMET-XL-QE	0.264	0.252	0.208	0.242
xCOMET-XXL	0.320	0.262	0.257	0.280
xCOMET-XXL-QE	0.298	0.238	0.229	0.255
EviSpan (Ours)	0.240	0.291	0.273	0.268
EviSpan-QE (Ours)	0.202	0.256	0.256	0.238

Table 2: Weighted character-level F1 scores ($\uparrow$) on error span detection for the WMT22 News-domain MQM subset, following the xCOMET evaluation setting.

6.1 Translation Scoring Performance

Table 1 shows that EviSpan obtains the best numerical average within the Generative/LLM Judge family on both WMT22 and WMT24 MQM. Compared with the closest same-size generative baseline, Remedy-R-14B, EviSpan achieves higher averages on both benchmarks while additionally producing tagged spans, severity labels, and rationales before the final score. Since paired significance tests are not uniformly significant across all levels and language pairs, we interpret these results as evidence that explicit span grounding can be integrated into generative MT evaluation while maintaining competitive scoring performance, rather than as a claim of scalar-metric dominance. Per-language scoring breakdowns for EviSpan and EviSpan-QE are provided in Appendix B.1.

6.2 Error Span Detection Accuracy

Table 2 evaluates generated error spans with weighted character-level F1 on the WMT22 News-domain MQM subset. EviSpan achieves an average span F1 of 0.268, comparable to xCOMET-XL (0.269) and higher than AutoMQM with GPT-4 prompting (0.262), though below xCOMET-XXL (0.280). This indicates that a generative evaluator can produce competitive localized evidence without a dedicated sequence-tagging head.

6.3 Feedback Usefulness Compared with Remedy-R

Table 3 reports pairwise LLM-as-a-judge results comparing EviSpan’s span-grounded feedback with Remedy-R’s free-form reasoning feedback. Across 900 WMT22 MQM examples and three judge models, EviSpan obtains more strict

Criterion	Win	Loss	Tie
Clarity	48.33 ± 1.30	18.96 ± 1.01	32.70 ± 1.13
Correctness	44.00 ± 1.37	19.96 ± 1.11	36.04 ± 1.28
Explanation qual.	42.48 ± 1.39	35.93 ± 1.31	21.59 ± 0.93
Key issue ID	37.41 ± 1.42	19.52 ± 1.14	43.07 ± 1.43
Overall use.	43.96 ± 1.38	32.41 ± 1.27	23.63 ± 0.97
Revision use.	39.22 ± 1.40	28.07 ± 1.21	32.70 ± 1.21

Table 3: Pairwise LLM-as-a-judge evaluation results. Values are mean percentages with bootstrap standard errors. Win, Loss, and Tie indicate whether EviSpan feedback, Remedy-R feedback, or neither was preferred.

preferences than Remedy-R on all six criteria, although many examples are judged as ties.

EviSpan shows the clearest advantage in clarity, correctness, and key issue identification. This pattern suggests that explicit span evidence makes the feedback easier to inspect and helps judges verify whether the explanation is grounded in the source, hypothesis, and reference. The comparison is closer for explanation quality and revision usefulness, where Remedy-R’s longer multi-aspect analyses may provide broader guidance.

We therefore interpret the result as a consistent but not absolute preference advantage: EviSpan’s strict wins consistently exceed strict losses, but ties remain common. Additional judge details, breakdowns, qualitative examples, and the supplementary xTower comparison are provided in Appendices C–D.

6.4 Reward and Curriculum Ablations

Table 4 examines how reward components and training schedules affect the trade-off between scalar evaluation accuracy and span-grounded evidence generation. The component ablations show that span and score rewards are complementary: each reward most strongly improves its direct target, while both also yield cross-objective gains. The curriculum ablations further show that optimization order matters. The final span-only $\rightarrow$ joint schedule gives the best overall average and the strongest Span F1, whereas reversed or one-stage joint training weakens either system-level accuracy or span grounding.

6.5 Does the Final Score Use the Exposed Evidence?

We test score sensitivity with an evidence-prefix intervention on WMT22 MQM. Instead of generating the full output, the model receives a fixed

Ablation	Training schedule	Avg.	WMT22			WMT24			Span F1
			Avg.	Acc	acc_{eq}^*	Avg.	SPA	acc_{eq}^*	
Prompting baseline	no training	70.7	71.6	89.1	54.1	69.8	83.7	55.9	0.160
w/o score reward	span only	72.1	72.4	89.1	55.6	71.8	86.5	57.2	0.245
w/o span reward	score only	73.2	73.5	89.5	57.4	72.9	87.1	58.8	0.188
w/o staged curriculum	joint only	72.9	72.9	89.3	56.5	72.9	87.8	57.9	0.242
reversed curriculum	score only $\rightarrow$ joint	73.2	73.5	88.1	58.8	72.8	86.2	59.3	0.257
EviSpan	span only $\rightarrow$ joint	73.4	73.6	90.2	57.0	73.2	88.0	58.5	0.268

Table 4: Ablation study on reward components and training curriculum. All models use the same backbone. When active, the span reward is computed with $F_{0.5}$. WMT24 reports system-level SPA and segment-level tie-calibrated accuracy, while WMT22 reports system-level pairwise accuracy and segment-level tie-calibrated accuracy. Span F1 is measured on the WMT22 News test set with weighted character-level span matching and is not included in the overall Avg.

Condition	Sys Acc	Seg acc^*	Clean score	Mod. score	Low score
Human MQM	–	–	99.2	92.0	68.4
Standard EviSpan	90.2	57.0	93.0	91.4	83.9
Gold MQM evid.	94.3	71.0	94.0	91.6	79.6
No-error evid.	89.7	54.6	94.3	93.6	87.2

Table 5: Evidence-prefix intervention on WMT22 MQM. Clean, Mod., and Low report mean scores for clean/high-quality, minor/moderate, and multiple/low-quality strata. The Human MQM row gives the mean gold score for each stratum.

assistant prefix containing the two local-evidence fields, `[[[annotated_translation]]]` and `[[[errors]]]`, and then generates only the rationale and final score. We compare standard EviSpan inference with a gold MQM prefix, which uses human span tags and error labels, and a no-error prefix, which copies the untagged hypothesis and sets the error list to `none`. The source, hypothesis, and reference are identical across conditions.

For score-behavior analysis, we stratify examples using a hybrid gold MQM criterion based on both human scores and annotated errors. Clean/high-quality examples have no annotated errors or have a gold score of at least 95 without major or critical errors; multiple/low-quality examples have at least three errors, a gold score of at most 80, or a critical error; the remaining errorful examples are grouped as minor/moderate.

Table 5 shows that exposed evidence is actionable for scoring. Compared with standard EviSpan inference, clamping gold MQM evidence increases WMT22 segment-level acc_{eq}^* from 57.0 to 71.0 and moves multiple/low-quality examples

closer to their human MQM score, from 83.9 to 79.6 toward 68.4. Conversely, the no-error prefix raises the same group to 87.2, indicating that replacing the local evidence shifts the final score in the expected direction. At the same time, low-quality examples remain below clean examples even under the no-error prefix, indicating that the final score is not determined by the evidence prefix alone.

7 Conclusion

We introduced EviSpan, an evidence-first generative evaluator for machine translation that marks MQM-style error spans before producing an error list, rationale, and final quality score. Trained with MQM-derived verifiable rewards and a two-stage GRPO curriculum, EviSpan remains competitive with the same-size Remedy-R generative baseline in WMT scoring while producing explicit span-level evidence. Ablations show that span and score rewards play complementary roles, and that stabilizing span grounding before joint optimization yields the best accuracy-grounding trade-off. Pairwise feedback evaluation further suggests that exposing explicit spans makes generative evaluation feedback easier to inspect and more useful than free-form reasoning alone. Finally, evidence-prefix interventions show that the exposed spans are actionable for scoring, although they do not fully determine the final score. Overall, EviSpan shows that explicit local grounding can be integrated into generative MT evaluation without giving up competitive scoring performance, providing a practical interface between scalar metrics, span taggers, and rationale-based evaluators.

Limitations

Causal faithfulness. Our evidence-prefix intervention demonstrates evidence sensitivity rather than mechanistic faithfulness. Gold MQM evidence improves scoring, and a no-error prefix shifts scores upward for errorful translations. However, the no-error prefix does not fully determine the final score, indicating that EviSpan also uses the source, hypothesis, and reference through a direct input-based scoring pathway. We therefore claim inspectable and actionable evidence use, not that the generated spans causally determine the score. Future work should test stronger counterfactual interventions, such as inconsistent span–score prefixes, targeted span deletion, and controlled evidence perturbations, to better separate evidence-mediated scoring from direct input-based scoring.

Domain and document-level coverage. Although EviSpan transfers to language pairs not seen during training, our experiments are still limited in domain coverage and input granularity. The main training data comes from WMT20 MQM annotations for English-German and Chinese-English, while evaluation is conducted on WMT MQM benchmarks. These results do not fully establish robustness for specialized domains, very low-resource language pairs, pathological translation outputs, or document-level translation. Long-form evaluation may require modeling discourse-level consistency, omissions, and cross-sentence dependencies beyond sentence-level span tags. Future work should evaluate span-grounded generative metrics on broader domains, challenge sets, and document-level benchmarks.

Limits of span-based evidence. Span-based evidence is not equally suitable for all translation errors. Omissions, discourse-level mistranslations, and pragmatic inconsistencies may not correspond to a single contiguous span in the translation hypothesis. In such cases, span tags should be viewed as partial local evidence rather than a complete explanation of the error. Moreover, if the predicted span is incorrect, the subsequent feedback can appear grounded while pointing to the wrong evidence. Future work should consider richer evidence representations, such as source-side spans, omission markers, non-contiguous spans, and document-level links, instead of relying only on contiguous target-side spans.

Error category supervision. EviSpan generates MQM-style error categories as part of its structured feedback, but our rewards do not directly optimize fine-grained category classification. They instead verify format validity, span localization, severity agreement, and score calibration. Diagnostic analyses suggest that generated labels can reflect category priors, especially toward frequent categories such as Accuracy/Mistranslation. Future work should explore category-aware rewards, label normalization, or constrained label spaces to improve the diagnostic reliability of generated MQM labels.

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A Implementation and Reproducibility Details

A.1 Notation

Table 6 summarizes the main notation used in the method, training objective, and evaluation protocols.

A.2 Prompt and Output Example

Table 7 shows the reference-based prompt template used for both training rollouts and inference, together with an actual model output. MQM annotations are used only for reward computation and are never shown in the prompt. The reference-free (QE) prompt uses the same system instruction, output schema, and decoding setup as the reference-based prompt, except that the reference line is omitted from the user prompt.

A.3 Training Hyperparameters and Compute

Table 8 summarizes the main training hyperparameters and compute setup used for EviSpan. Both stages are optimized with GRPO using the same WMT20 MQM training data and the same output format. Stage 1 optimizes format-valid span grounding, while Stage 2 initializes from the selected Stage-1 checkpoint and adds score calibration. All reported main results use the final Stage-2 checkpoint selected by validation reward. We choose δ and D to balance sensitivity and stability: δ should be large enough to avoid overly sharp rewards but small enough to distinguish meaningful score errors, while D prevents far-off predictions from receiving substantial reward.

Compute budget. The final EviSpan training run required approximately 80 NVIDIA H100 GPU-hours. Training was conducted on 4-8 NVIDIA H100 GPUs, with wall-clock time ranging from about 10 to 20 hours depending on the distributed configuration.

A.4 Parsing Reliability

Table 9 reports structured-output parsing reliability on the WMT22 and WMT24 MQM test sets. Parsing failures are rare in both benchmarks. We nevertheless retain all failed outputs in evaluation and apply the conservative fallback described in Section 5.3: unparseable scores are assigned 0, and unparseable span fields are treated as predicting no error spans.

Notation	Description
<i>Input and output</i>	
src	Source sentence.
ref_{opt}	Optional reference translation; omitted in the reference-free setting.
hyp	Translation hypothesis to be evaluated.
$inst$	Task instruction specifying the evaluation criteria and output format.
x	Input prompt consisting of the instruction, source, optional reference, and hypothesis.
y	Structured model output containing tagged spans, error labels, rationale, and score.
$\hat{E}$	Predicted error-span set extracted from the generated output.
$\hat{s}$	Predicted segment-level quality score.
<i>MQM annotations</i>	
$\mathcal{M}(x)$	Retained MQM annotations for input x .
J_x	Number of retained annotators for input x .
E_j^*	Gold error-span set from annotator j .
s_j^*	Gold MQM-derived quality score from annotator j .
$E^*(x)$	Collection of retained gold error-span annotations.
$s^*(x)$	Annotator-averaged gold quality score.
<i>Rewards</i>	
g_{fmt}	Binary format-validity gate.
r_{span}	Span-grounding reward based on character-level overlap with MQM spans.
r_{score}	Score-calibration reward based on the predicted and gold quality scores.
r_{task}	Weighted task reward before format gating.
r	Final reward used for GRPO optimization.
λ_{span}	Weight for the span-grounding reward.
λ_{score}	Weight for the score-calibration reward.
F_{β}^{char}	Character-level span-overlap score used for span reward and evaluation.
D	Maximum clipped score difference in the score reward.
δ	Huber-style transition threshold in the score reward.
<i>GRPO optimization</i>	
π_{θ}	Trainable rollout policy.
π_{ref}	Frozen reference policy for KL regularization.
G	Number of sampled outputs per input.
y_i	The i -th sampled structured output.
r_i	Reward assigned to the i -th sampled output.
A_i	Group-relative advantage for the i -th sampled output.
μ_r, σ_r	Mean and standard deviation of rewards within a rollout group.
β_{KL}	KL regularization coefficient.
<i>Evaluation metrics</i>	
Acc	System-level pairwise accuracy.
SPA	System-level Soft Pairwise Accuracy.
acc_{eq}^*	Segment-level tie-calibrated pairwise accuracy.

Table 6: Notation used throughout the paper. Hatted variables denote model predictions, and starred variables denote MQM-derived gold annotations.

System prompt

```
You are an expert machine translation quality evaluator.
Identify translation errors and assess quality using the following core dimensions.

### Error Categories:
- Accuracy (e.g., mistranslation, omission, addition)
- Fluency (e.g., grammar, spelling, punctuation)
- Style (e.g., awkward phrasing, register)
- Terminology (e.g., inconsistent use, inappropriate context)

### Severity Levels:
- Minor: Slight degradation, core meaning intact.
- Major: Core meaning is altered or obscured.

### Output Format:
Output your evaluation using the section headers below, each on its own line.

[[[annotated_translation]]]
Copy the original Translation exactly verbatim, inserting <vN></vN> tags around errors (e.g., <v0>first</v0>, <v1>second</v1>). You MUST NOT alter capitalization, punctuation, spelling, spacing, or add any new words. Only insert the tags. If there are no errors, output the Translation exactly as is.

[[[errors]]]
List each error on a separate line in the format: vN | category | severity. category describes the error type, formatted as Category/Subcategory (e.g. Accuracy/Mistranslation). severity is one of Major or Minor. If there are no errors, write "none"

[[[rationale]]]
Based on your analysis of the identified errors, explain the brief reasons behind the assessment in English.

[[[score]]]
A quality score (0-100). Deduct logically based on the rationale and the errors. Output only the integer score without any additional text.
```

User prompt

```
Source (German): Er kam am Dienstag an, vergaß jedoch, die Unterlagen mitzubringen.
Reference (may be imperfect): He arrived on Tuesday, but forgot to bring the documents.
Translation (English): He arrive on Monday, but forgot to bring the document.
```

Model output

```
[[[annotated_translation]]]
He <v0>arrive</v0> on <v1>Monday</v1>, but forgot to bring the <v2>document</v2>.

[[[errors]]]
v0 | Accuracy/Mistranslation | Minor
v1 | Accuracy/Mistranslation | Major
v2 | Accuracy/Omission | Minor

[[[rationale]]]
The translation contains minor errors in verb conjugation ("arrive" instead of "arrived") and a significant error in the day mentioned ("Monday" instead of "Tuesday"). Additionally, there is a minor error in the use of "document" instead of "documents," which is a plural noun in the source text.

[[[score]]]
80
```

Table 7: Example input prompt and an actual EviSpan output for reference-based MT evaluation. The source is German, and the hypothesis is an English translation.

Hyperparameter	Value
Backbone	Qwen2.5-Instruct
Main model	Qwen2.5-14B-Instruct
Fine-tuning method	LoRA
LoRA rank / alpha	64 / 128
Learning rate	2×10^{-5}
Warmup ratio	0.1
GRPO rollouts per prompt	8
Sampling temperature	0.8
KL coefficient	0.04
Per-device batch size	2
Effective batch size	256
Training length	up to 1 epoch per stage
Stage 1 reward weights	$\lambda_{\text{span}} = 1, \lambda_{\text{score}} = 0$
Stage 2 reward weights	$\lambda_{\text{span}} = 1, \lambda_{\text{score}} = 0.2$
Reward hyperparameters	$\beta = 0.5, D = 30, \delta = 10$

Table 8: Main training hyperparameters.

Benchmark	Outputs	Parsed	Failed	Success rate
WMT22	106,758	106,741	17	99.984%
WMT24	68,502	68,326	176	99.743%
Total	175,260	175,067	193	99.890%

Table 9: Parsing reliability of EviSpan outputs on the WMT22 and WMT24 MQM test sets. A failure means that the generated output could not be parsed by the deterministic evaluator under the required section and score format.

Method	Level	All	En-De	En-Ru	Zh-En
EviSpan	Sys Acc	90.2	—	—	—
EviSpan	Seg acc_{eq}^*	57.0	58.1	57.8	55.0
EviSpan-QE	Sys Acc	90.1	—	—	—
EviSpan-QE	Seg acc_{eq}^*	55.8	57.6	55.9	53.9

(a) WMT22 MQM

Method	Level	All	En-De	En-Es	Ja-Zh
EviSpan	Sys SPA	88.0	85.8	84.7	93.4
EviSpan	Seg acc_{eq}^*	58.5	51.0	68.0	56.3
EviSpan-QE	Sys SPA	87.6	85.5	84.6	92.6
EviSpan-QE	Seg acc_{eq}^*	58.3	51.1	68.1	55.6

(b) WMT24 MQM

Table 10: Per-language scoring breakdown for EviSpan and EviSpan-QE. For WMT22, Sys Acc is reported only as the official aggregate over all language pairs, matching Table 1; per-language columns are therefore shown only for segment-level acc_{eq}^* . For WMT24, Sys SPA and segment-level acc_{eq}^* are reported by language pair.

B Additional Results

B.1 Per-language Scoring Breakdown

Table 10 reports additional scoring breakdowns for EviSpan and EviSpan-QE. These results follow the same repeated inference/evaluation averaging protocol as Table 1. For WMT22, we report the official aggregate system-level Acc rather than averaging language-pair-level system accuracies.

B.2 Effect of Supervised Warm-up

Table 11 examines whether supervised warm-up provides a useful cold start before GRPO. Because gold rationales are unavailable, the SFT stage is trained on rationale-free serialized MQM outputs, while the final evaluator is expected to generate rationale-inclusive feedback before assigning a score. SFT alone slightly improves scoring over prompting but does not improve span grounding. After span-focused GRPO, SFT warm-up recovers strong segment-level accuracy, but its Span F1 remains similar to direct span-only GRPO.

The full SFT $\rightarrow$ span only $\rightarrow$ joint schedule presents a sharper trade-off. It obtains the highest Span F1 and the best segment-level accuracies, suggesting improved local evidence matching and segment-level discrimination. However, its system-level WMT22 acc_{eq}^* and WMT24 SPA drop substantially. This pattern suggests that SFT warm-up may induce a brittle scoring prior: subsequent joint GRPO improves local span matching, but does not preserve the calibrated score scale needed for system-level ranking. We therefore use direct span-first GRPO for EviSpan, which gives the best overall scoring-grounding trade-off.

C LLM-as-a-Judge Details for Remedy-R Comparison

This appendix provides additional details for the rationale-usefulness evaluation described in Section 5.4. The evaluation compares feed-

back generated by our span-grounded evaluator against Remedy-R feedback using three independent LLM judges: Gemini-2.5-Pro (Comanici et al., 2025), GLM-5.1 (Zeng et al., 2026), and DeepSeek-R1 (Guo et al., 2025). Unlike WMT meta-evaluation and span-F1, this analysis does not evaluate final scalar scores or span overlap directly, but instead measures whether the generated feedback is useful, grounded and actionable for translation review.

C.1 Sampling and Compared Outputs

We sample examples from the WMT22 MQM test set for En-De, En-Ru, and Zh-En. To avoid evaluating only easy or only severe cases, we use quality-stratified sampling over three groups: clean or high-quality translations, translations with minor or moderate errors and low-quality translations with multiple or severe errors. For each sampled example, we generate feedback from our model and from Remedy-R using the same source sentence, translation hypothesis, and reference.

Before presenting the outputs to the judge, we remove final scalar scores from both candidates. This prevents the judge from preferring a feedback solely because its numeric score appears closer to the expected quality. The compared content therefore consists only of the explanation parts of each output: for our model, the tagged translation, error list, and rationale; and for Remedy-R, its natural-language evaluation analysis.

C.2 Judge Prompt and Output Schema

Listing 1 provides the prompt and required output schema for the pairwise LLM-as-a-judge evaluation. Candidate identities are hidden, and the two feedbacks are randomly assigned to A and B.

Variant	Training schedule	Avg.	WMT22			WMT24			Span F1
			Avg.	Acc	acc_{eq}^*	Avg.	SPA	acc_{eq}^*	
Prompting baseline	no training	70.7	71.6	89.1	54.1	69.8	83.7	55.9	0.160
SFT warm-up	SFT	71.2	71.6	89.7	53.4	70.9	85.9	55.8	0.147
SFT warm-up	SFT $\rightarrow$ span	72.7	73.3	89.1	57.5	72.2	86.2	58.3	0.247
SFT warm-up	SFT $\rightarrow$ span $\rightarrow$ joint	72.3	72.9	87.5	58.3	71.7	84.7	58.7	0.299
Direct GRPO	span only	72.1	72.4	89.1	55.6	71.8	86.5	57.2	0.245
EviSpan	span $\rightarrow$ joint	73.4	73.6	90.2	57.0	73.2	88.0	58.5	0.268

Table 11: Effect of supervised warm-up before GRPO. Because no gold rationales are available, SFT warm-up is trained with rationale fields removed from both the prompt and target output. Although SFT $\rightarrow$ span $\rightarrow$ joint achieves the highest Span F1, it substantially reduces system-level accuracy on both WMT22 and WMT24. Direct span-first GRPO therefore provides a better scoring-grounding trade-off.

```

[System]
You are a strict bilingual machine translation evaluation judge.

Your task is to compare two anonymous evaluation feedbacks for the same machine translation.
Judge which feedback would be more useful for a human reviewer who wants to understand whether the
translation is good, what problems it has, and how it might be revised.

Important rules:
- Do NOT favor a candidate because of its format, structure, tags, section headings, or length.
- Do NOT penalize a candidate for lacking explicit tags or MQM-style labels.
- Do NOT reward verbosity. A longer feedback is better only if it adds correct and useful information.
- Do NOT reward brevity by itself. A shorter feedback is better only if it remains correct and useful.
- Every evaluative claim must be supported by the Source, Translation, and Reference if provided.
- The Reference may be imperfect. Use it as helpful evidence, but the Source is authoritative.
- A good feedback should correctly identify important translation problems, avoid inventing problems,
explain the issue clearly, and help a reviewer decide whether or how to revise the translation.
- If both feedbacks are similarly useful or similarly flawed, choose "Tie".
- Return ONLY a valid JSON object.

[User]
We are evaluating a machine translation.

Language pair: {LANG_PAIR}

Source: {SOURCE}

Reference: {REFERENCE}

Translation: {TRANSLATION}

Candidate A feedback: {FEEDBACK_A}

Candidate B feedback: {FEEDBACK_B}

Compare Candidate A and Candidate B using the criteria below.

Criteria:
1. Correctness:
  Are the feedback's claims accurate and supported by the Source, Translation, and Reference if provided?
  Penalize invented errors, unsupported claims, incorrect explanations, incorrect quotations, or
  contradictions.

2. Key issue identification:
  Does the feedback identify the most important translation issue(s), if any?
  It should not miss major problems, but it should also avoid inventing problems in good translations.

3. Explanation quality:
  Does the feedback explain why the issue matters for meaning, fluency, completeness, or translation
  quality?
  The explanation should help the reviewer understand the translation quality, not merely repeat labels.

4. Revision usefulness:
  Would the feedback help a translator, post-editor, or reviewer decide whether and how to revise the
  translation?

5. Clarity:
  Is the feedback easy to understand and not misleading or distracting?
  Do not judge clarity based only on length, formatting, tags, or section structure.

Use a 0-4 scale for each criterion:
0 = mostly wrong or unsupported
1 = weak
2 = partially useful
3 = mostly useful
4 = excellent

Return ONLY this JSON format:
{{
  "scores": {{
    "A": {{
      "correctness": 0,
      "key_issue_identification": 0,
      "explanation_quality": 0,
      "revision_usefulness": 0,
      "clarity": 0,
      "overall": 0
    }},
    "B": {{
      "correctness": 0,
      "key_issue_identification": 0,
      "explanation_quality": 0,
      "revision_usefulness": 0,
      "clarity": 0,
      "overall": 0
    }}
  }}
}}

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    },
    "pairwise": {
      "correctness": "A|B|Tie",
      "key_issue_identification": "A|B|Tie",
      "explanation_quality": "A|B|Tie",
      "revision_usefulness": "A|B|Tie",
      "clarity": "A|B|Tie",
      "overall_usefulness": "A|B|Tie"
    },
    "winner": "A|B|Tie",
    "confidence": 1,
    "brief_reason": "In at most 60 words, explain the most important difference.",
    "unsupported_or_wrong_claims": {
      "A": [],
      "B": []
    },
    "important_missed_errors": {
      "A": [],
      "B": []
    }
  }
}

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Listing 1: Prompt used for the pairwise LLM-as-a-judge evaluation.

Criterion	Gemini-2.5-Pro				GLM-5.1				DeepSeek-R1			
	Win	Loss	Tie	Δ	Win	Loss	Tie	Δ	Win	Loss	Tie	Δ
Correctness	40.89 $\pm$ 1.66	22.78 $\pm$ 1.41	36.33 $\pm$ 1.62	+18.11 $\pm$ 2.63	49.33 $\pm$ 1.67	19.22 $\pm$ 1.30	31.44 $\pm$ 1.54	+30.11 $\pm$ 2.57	41.78 $\pm$ 1.65	17.89 $\pm$ 1.29	40.33 $\pm$ 1.64	+23.89 $\pm$ 2.46
Key issue ID	34.89 $\pm$ 1.59	22.44 $\pm$ 1.39	42.67 $\pm$ 1.63	+12.44 $\pm$ 2.50	39.00 $\pm$ 1.62	18.44 $\pm$ 1.30	42.56 $\pm$ 1.65	+20.56 $\pm$ 2.43	38.33 $\pm$ 1.63	17.67 $\pm$ 1.27	44.00 $\pm$ 1.66	+20.67 $\pm$ 2.41
Explanation quality	41.11 $\pm$ 1.65	36.00 $\pm$ 1.60	22.89 $\pm$ 1.40	+5.11 $\pm$ 2.93	47.11 $\pm$ 1.68	32.00 $\pm$ 1.57	20.89 $\pm$ 1.36	+15.11 $\pm$ 2.95	39.22 $\pm$ 1.62	39.78 $\pm$ 1.63	21.00 $\pm$ 1.34	-0.56 $\pm$ 2.96
Revision usefulness	36.00 $\pm$ 1.61	26.89 $\pm$ 1.47	37.11 $\pm$ 1.61	+9.11 $\pm$ 2.62	43.44 $\pm$ 1.66	25.22 $\pm$ 1.45	31.33 $\pm$ 1.53	+18.22 $\pm$ 2.71	38.22 $\pm$ 1.61	32.11 $\pm$ 1.57	29.67 $\pm$ 1.54	+6.11 $\pm$ 2.78
Clarity	50.22 $\pm$ 1.67	20.89 $\pm$ 1.36	28.89 $\pm$ 1.50	+29.33 $\pm$ 2.65	52.33 $\pm$ 1.67	14.00 $\pm$ 1.15	33.67 $\pm$ 1.56	+38.33 $\pm$ 2.40	42.44 $\pm$ 1.64	22.00 $\pm$ 1.39	35.56 $\pm$ 1.61	+20.44 $\pm$ 2.58
Overall usefulness	41.67 $\pm$ 1.64	35.44 $\pm$ 1.60	22.89 $\pm$ 1.39	+6.22 $\pm$ 2.93	50.33 $\pm$ 1.66	26.56 $\pm$ 1.47	23.11 $\pm$ 1.41	+23.78 $\pm$ 2.80	39.89 $\pm$ 1.63	35.22 $\pm$ 1.60	24.89 $\pm$ 1.42	+4.67 $\pm$ 2.90

Table 12: Judge-model breakdown of the pairwise LLM-as-a-judge comparison between EviSpan feedback and Remedy-R feedback. Values are reported as mean $\pm$ standard error. Win, Loss, and Tie are percentages for EviSpan against Remedy-R after mapping randomized A/B decisions back to model identities. Δ denotes the win-loss difference in percentage points. For each judge model, standard errors are estimated by bootstrap resampling over A/B evaluation pairs after aggregating the original and swapped judgments for each pair.

We use only the preference labels for the reported win/loss/tie analyses. The remaining fields are collected only for manual inspection and are not used for aggregation.

C.3 Additional Analysis

Table 13 reports inter-judge agreement among the three LLM judges. The judges show moderate agreement overall, with Fleiss’ κ ranging from 0.367 to 0.626 across criteria. Agreement is highest for key issue identification, suggesting that judges are relatively consistent in deciding whether a feedback captures the main translation issue. Clarity has the lowest agreement, which is expected because readability judgments are more subjective than factual correctness or issue identification.

Table 14 breaks down the results by translation quality stratum. The preference for our feedback becomes stronger as translation quality decreases. For overall usefulness, the tie-inclusive win rate is close to neutral for clean or high-quality translations, but increases for translations with minor or moderate errors and is highest for multiple-error or low-quality translations. This pattern is consistent with the intended diagnostic role of span-grounded feedback: explicit local evidence may be especially helpful when reviewers need to locate and understand translation errors.

The A/B swap analysis in Table 15 shows that hard flips are relatively rare, ranging from 4.6% for clarity to 11.3% for overall usefulness. Most unstable cases involve a strict preference in one order and a tie in the other, rather than opposite strict preferences. This suggests that the aggregate preference trends are not primarily driven by severe position-order reversals, although some criteria, especially overall usefulness and explanation quality, remain more sensitive to judge uncertainty.

Criterion	Fleiss’ κ	Observed agreement	Unanimous
Correctness	0.511	0.689	0.557
Key issue ID	0.626	0.762	0.661
Explanation	0.451	0.647	0.504
Revision use	0.490	0.663	0.518
Clarity	0.367	0.605	0.446
Overall	0.425	0.629	0.477

Table 13: Inter-judge agreement among the three LLM judges. Observed agreement is computed over three labels: our feedback preferred, Remedy-R feedback preferred, and tie. Unanimous denotes the fraction of examples where all three judges assign the same label.

D Additional Comparison with xTower

This appendix reports an additional comparison with xTower. Unlike the Remedy-R comparison in Appendix C, this setting is not an explanation-only comparison. xTower receives external xCOMET-XL marked spans as input and may output explicit translation corrections in addition to error explanations, whereas EviSpan jointly generates spans, error labels, and rationales. This makes xTower a particularly challenging feedback baseline, since it is given external localized evidence and is optimized for correction-oriented feedback. We therefore interpret this experiment as a supplementary comparison with a strong correction-oriented feedback baseline rather than as a direct rationale-only comparison.

For consistency, we use the same judge prompt, JSON schema, criteria, A/B randomization, swapped-order evaluation, and aggregation procedure as in the Remedy-R comparison. The only change is the baseline being compared.

D.1 Main Results

Table 16 reports the pairwise comparison between EviSpan feedback and xTower feedback. Overall, EviSpan receives more strict preferences than xTower on five of the six criteria. The largest mar-

Criterion	Win	Loss	Tie	Δ_{W-L}
Clean/high				
Clarity	40.7 $\pm$ 2.2	15.7 $\pm$ 1.6	43.7 $\pm$ 2.2	+25.0 $\pm$ 3.2
Correctness	35.4 $\pm$ 2.3	14.8 $\pm$ 1.7	49.8 $\pm$ 2.4	+20.7 $\pm$ 3.3
Explanation quality	34.7 $\pm$ 2.3	40.3 $\pm$ 2.3	25.0 $\pm$ 1.7	-5.7 $\pm$ 4.3
Key issue ID	28.1 $\pm$ 2.3	11.7 $\pm$ 1.7	60.2 $\pm$ 2.5	+16.4 $\pm$ 3.2
Overall usefulness	37.3 $\pm$ 2.4	35.9 $\pm$ 2.2	26.8 $\pm$ 1.7	+1.4 $\pm$ 4.3
Revision usefulness	31.4 $\pm$ 2.3	25.9 $\pm$ 2.0	42.7 $\pm$ 2.2	+5.6 $\pm$ 3.7
Minor/moderate				
Clarity	51.6 $\pm$ 2.3	21.7 $\pm$ 1.8	26.8 $\pm$ 1.7	+29.9 $\pm$ 3.7
Correctness	48.2 $\pm$ 2.3	22.3 $\pm$ 2.0	29.4 $\pm$ 2.0	+25.9 $\pm$ 3.8
Explanation quality	44.2 $\pm$ 2.4	35.4 $\pm$ 2.3	20.3 $\pm$ 1.5	+8.8 $\pm$ 4.4
Key issue ID	40.6 $\pm$ 2.4	24.7 $\pm$ 2.1	34.8 $\pm$ 2.3	+15.9 $\pm$ 3.9
Overall usefulness	46.1 $\pm$ 2.3	31.9 $\pm$ 2.1	22.0 $\pm$ 1.6	+14.2 $\pm$ 4.2
Revision usefulness	41.0 $\pm$ 2.4	29.3 $\pm$ 2.1	29.7 $\pm$ 2.0	+11.7 $\pm$ 4.1
Multiple/low				
Clarity	52.8 $\pm$ 2.2	19.6 $\pm$ 1.8	27.7 $\pm$ 1.8	+33.2 $\pm$ 3.6
Correctness	48.3 $\pm$ 2.4	22.8 $\pm$ 2.0	28.9 $\pm$ 2.0	+25.6 $\pm$ 3.9
Explanation quality	48.6 $\pm$ 2.4	32.0 $\pm$ 2.3	19.4 $\pm$ 1.6	+16.6 $\pm$ 4.4
Key issue ID	43.6 $\pm$ 2.4	22.2 $\pm$ 2.0	34.2 $\pm$ 2.3	+21.3 $\pm$ 3.8
Overall usefulness	48.4 $\pm$ 2.4	29.4 $\pm$ 2.2	22.1 $\pm$ 1.7	+19.0 $\pm$ 4.2
Revision usefulness	45.2 $\pm$ 2.4	29.0 $\pm$ 2.2	25.8 $\pm$ 1.9	+16.2 $\pm$ 4.2

Table 14: Pairwise LLM-as-a-judge results by translation quality stratum. Values are reported as mean $\pm$ standard error. Standard errors are estimated by bootstrap resampling over prompt pairs, preserving the three judge-model outcomes associated with each prompt. Δ_{W-L} denotes the win-loss difference in percentage points. Positive values indicate a preference for EviSpan.

Criterion	Win/Win	Loss/Loss	Tie/Tie	Win/Tie	Loss/Tie	Split	Instability
Clarity	30.4	10.3	28.1	17.9	8.6	4.6	31.1
Correctness	34.1	14.7	29.5	9.9	5.3	6.5	21.7
Explanation quality	33.9	23.6	13.5	8.6	12.3	8.1	29.0
Key issue ID	29.3	14.8	38.1	8.1	4.7	5.0	17.8
Overall usefulness	36.9	24.0	12.4	7.1	8.4	11.3	26.8
Revision usefulness	31.6	19.2	25.1	7.6	8.9	7.6	24.1

Table 15: A/B swap consistency analysis for the pairwise LLM-as-a-judge evaluation. Each row contains 2,700 paired decisions, where the same prompt pair is evaluated under both A/B orders. Win/Win, Loss/Loss, and Tie/Tie denote order-invariant decisions. Win/Tie and Loss/Tie denote cases where one order yields a strict preference and the swapped order yields a tie. Split denotes hard flips, where the two A/B orders yield opposite strict preferences. Instability is the sum of Win/Tie, Loss/Tie, and Split. All values are percentages.

gins are observed for clarity, where EviSpan wins 56.3% of judgments and loses 26.1%, and correctness, where it wins 51.1% and loses 31.4%. EviSpan also obtains smaller positive margins for key issue identification, explanation quality, and overall usefulness.

The only criterion where xTower has a slight advantage is revision usefulness, where EviSpan wins 39.3% and loses 42.0%. This result is unsurprising because xTower explicitly includes correction-oriented output, whereas EviSpan is designed primarily as an evaluator that exposes span-grounded evidence and a concise rationale. We therefore treat these results as supplementary evidence that EviSpan remains competitive against a correction-inclusive baseline with exter-

nal span input, rather than as the main feedback comparison.

D.2 Agreement and Diagnostics

Table 17 reports inter-judge agreement for the xTower comparison. Agreement is moderate across criteria, with Fleiss’ κ ranging from 0.445 to 0.492 and observed agreement ranging from 0.650 to 0.694. This is reasonable for a rubric-based feedback comparison, especially because the candidates differ not only in explanation style but also in intended function.

The moderate κ values indicate that small margins should not be overinterpreted. For this reason, we focus on broad trends: EviSpan is clearly preferred for clarity and correctness, remains compet-

Criterion	All Judges				Gemini-2.5-Pro			GLM-5.1			DeepSeek-R1		
	Win	Loss	Tie	Δ_{W-L}	Win	Loss	Tie	Win	Loss	Tie	Win	Loss	Tie
Correctness	51.1 (1.38)	31.4 (1.30)	17.5 (0.90)	+19.6 (2.53)	51.4 (1.68)	29.3 (1.54)	19.2 (1.32)	56.6 (1.66)	30.8 (1.53)	12.7 (1.09)	45.2 (1.64)	34.2 (1.58)	20.6 (1.35)
Key issue ID	42.3 (1.39)	37.4 (1.35)	20.3 (0.96)	+4.9 (2.56)	44.1 (1.65)	34.2 (1.59)	21.7 (1.37)	46.0 (1.65)	36.7 (1.61)	17.3 (1.26)	36.9 (1.60)	41.3 (1.63)	21.8 (1.38)
Explanation quality	45.0 (1.37)	39.1 (1.35)	15.9 (0.82)	+5.9 (2.59)	46.6 (1.68)	35.7 (1.61)	17.8 (1.27)	51.3 (1.67)	36.0 (1.59)	12.7 (1.12)	37.1 (1.61)	45.7 (1.66)	17.2 (1.28)
Revision usefulness	39.3 (1.35)	42.0 (1.34)	18.7 (0.90)	-2.6 (2.54)	41.2 (1.64)	38.9 (1.63)	19.9 (1.33)	43.1 (1.65)	40.3 (1.63)	16.6 (1.23)	33.7 (1.58)	46.7 (1.65)	19.7 (1.33)
Clarity	56.3 (1.39)	26.1 (1.19)	17.6 (0.92)	+30.2 (2.42)	59.8 (1.64)	24.9 (1.45)	15.3 (1.20)	60.7 (1.62)	20.6 (1.34)	18.8 (1.30)	48.4 (1.68)	32.8 (1.57)	18.8 (1.31)
Overall usefulness	43.4 (1.40)	38.5 (1.34)	18.1 (0.84)	+4.9 (2.61)	43.4 (1.65)	37.1 (1.60)	19.4 (1.32)	49.0 (1.67)	35.7 (1.59)	15.3 (1.21)	37.8 (1.61)	42.8 (1.65)	19.4 (1.31)

Table 16: Pairwise LLM-as-a-judge comparison between EviSpan feedback and xTower feedback. Values are percentages for EviSpan against xTower after mapping randomized A/B decisions back to model identities, with standard errors shown in parentheses. The pooled block aggregates three judge models and swapped A/B orders. Win and Loss denote strict preferences for EviSpan and xTower, respectively; Tie denotes no clear preference. Δ_{W-L} is reported only for the pooled result and is computed as Win minus Loss. Standard errors were estimated with 10,000 prompt-level bootstrap resamples. For All judges, each resampled prompt retained the paired DeepSeek, Gemini, and GLM judge labels; for individual judges, resampling was performed over that judge’s prompt-level labels only.

Criterion	Fleiss’ κ	Observed agreement	Unanimous
Correctness	0.492	0.690	0.567
Key issue ID	0.491	0.674	0.547
Explanation	0.453	0.661	0.526
Revision use	0.456	0.655	0.518
Clarity	0.476	0.694	0.576
Overall	0.445	0.650	0.514

Table 17: Inter-judge agreement among the three LLM judges for the xTower comparison. Observed agreement is computed over three labels: EviSpan feedback preferred, xTower feedback preferred, and tie. Unanimous denotes the fraction of examples where all three judges assign the same label.

itive for overall usefulness, and is slightly behind xTower for revision usefulness.

E Qualitative Examples

This appendix provides selected qualitative examples comparing EviSpan and Remedy-R feedback on Chinese–English WMT22 MQM examples. Final scalar scores are omitted so that the comparison focuses on diagnostic feedback rather than numeric calibration. Red highlighting is added by the authors to indicate the target-side error span or repeated mentions of the same issue within the feedback. These examples are illustrative and are not used for quantitative aggregation.

The examples cover four representative cases: structural mistranslation (Table 18), a clean translation (Table 19), literal interpretation of a platform name (Table 20), and literal rendering of an idiom (Table 21).

Source	新的一年，第一件事是总结2021 调整2022战术布局。
Reference	In the new year, the first task is to summarize 2021 and adjust the tactical layout of 2022.
Translation	In the new year, the first thing is to summarize the 2021 adjustment of the 2022 tactical layout.
Remedy-R	Let's evaluate the translation step by step based on the given criteria: 1. Accuracy: - The source sentence "新的一年，第一件事是总结2021 调整2022战术布局。" translates to "In the new year, the first thing is to summarize 2021 and adjust the 2022 tactical layout." - The translation "In the new year, the first thing is to summarize the 2021 adjustment of the 2022 tactical layout." is somewhat accurate but not entirely clear. 调整 (adjust) is used twice, which is redundant and confusing. 总结2021 should be "summarize 2021" and 调整2022战术布局 should be "adjust the 2022 tactical layout." 2. Fluency: - The translation is somewhat awkward and not very natural. "Adjustment of the 2022 tactical layout" is not a common phrase in English. A more natural phrasing would be "tactical layout for 2022." "Summarize the 2021 adjustment" is also awkward. "Summarize 2021" is more natural. 3. Completeness: - The translation attempts to convey the meaning but is not entirely clear. 调整 is used twice, which is redundant and confusing. 总结2021 should be "summarize 2021" and 调整2022战术布局 should be "adjust the 2022 tactical layout." 4. Language Correctness: - The translation is mostly correct but awkward. "Adjustment" is used twice, which is not natural in English. "Summarize" and "adjust" should be used separately. Overall, the translation is not very clear and natural. It is somewhat accurate but not entirely so.
EviSpan	[Annotated Translation] In the new year, the first thing is to summarize the <v0>2021 adjustment of the 2022</v0> tactical layout. [Errors] v0 Accuracy/Mistranslation Major [Rationale] The translation incorrectly combines "总结2021" (summarize 2021) and "调整2022战术布局" (adjust the tactical layout of 2022) into a single phrase "summarize the 2021 adjustment of the 2022 tactical layout," which alters the core meaning of the original sentence. The correct translation should separate these actions, as shown in the reference.

Table 18: Qualitative example where Remedy-R repeatedly discusses the same structural mistranslation across multiple criteria. EviSpan localizes the phrase responsible for the altered predicate structure and gives a concise rationale grounded in that span. Red highlighting is added by the authors.

Source	丁子霖：致友人。
Reference	Ding Zilin: to friends.
Translation	Ding Zilin: to friends.
Remedy-R	Let's evaluate the translation step by step based on the given criteria: 1. Accuracy: - The source text "丁子霖：致友人。" translates to "Ding Zilin: to friends." "丁子霖" is correctly transliterated as "Ding Zilin." "致友人" is translated as "to friends," which is a literal translation but might not fully capture the nuance. "致" can also mean "dedicated to" or "addressed to," which might be more appropriate in some contexts. 2. Fluency: - The translation is mostly fluent and grammatically correct. "to friends" is a common and natural way to express "致友人" in English. - However, "to friends" might be more appropriate if it's part of a greeting or dedication, but it could be more natural to say "to my friends" or "to our friends" depending on the context. 3. Completeness: - The translation is complete in conveying the basic meaning of the source text. - However, "致" could be interpreted more contextually, such as "dedicated to" or "addressed to," which might be more appropriate in some contexts. 4. Language Correctness: - The translation is in correct English without any language mixing. - There are no grammatical errors or language correctness issues. 5. Adversarial and Noisy Input: - The source and translation are relevant and not empty or irrelevant. 6. Untranslated Text: - The translation is not an untranslated copy of the source text. Given the above evaluation, the translation is mostly accurate and fluent, but there is room for improvement in nuance and completeness.
EviSpan	[Annotated Translation] Ding Zilin: to friends. [Errors] none [Rationale] The translation is accurate, fluent, and maintains the appropriate style and terminology. There are no errors in the translation that affect accuracy, fluency, style, terminology, or locale conventions.

Table 19: Qualitative example where the translation closely matches the reference. Remedy-R proposes possible nuance-based revisions despite the absence of a clear error, whereas EviSpan predicts no error. Red highlighting is added by the authors.

Source	小雨伞买保险，靠谱吗？
Reference	Is it safe to buy insurance on Little Umbrella?
Translation	Is it reliable to buy insurance for a small umbrella?
Remedy-R	Let's evaluate the translation step by step based on the given criteria: 1. Accuracy: – The translation accurately conveys the meaning of the source text. 小雨伞 (xiǎoyǔsǎn) means "small umbrella," 买保险 (mǎi bǎoxiǎn) means "buy insurance," and 靠谱吗 (kǎopǔ ma) means "reliable?" The translation correctly captures the essence of the question. 2. Fluency: – The translation is fluent and reads naturally in English. The sentence structure is correct and the use of words is appropriate for a question in English. 3. Completeness: – All information from the source is conveyed accurately. The translation does not omit any part of the source text and maintains the same inquiry. 4. Language Correctness: – The translation is in correct English without any language mixing. The grammar and syntax are correct. 5. Adversarial and Noisy Input: – The source and translation are relevant and not empty or irrelevant. Given the above evaluation, the translation is of high quality.
EviSpan	[Annotated Translation] Is it reliable to buy insurance <v0>for a small umbrella</v0>? [Errors] v0 Accuracy/Mistranslation Major [Rationale] The translation incorrectly interprets "小雨伞" as "a small umbrella" instead of referring to the brand or platform "Little Umbrella". This mistranslation changes the core meaning of the sentence, making it a major error.

Table 20: Qualitative example where Remedy-R accepts a literal interpretation that is inconsistent with the reference and insurance context. EviSpan marks the local phrase that turns the platform name into a literal object. Red highlighting is added by the authors.

Source	(这对于投保人来说，就又是一个坑) 免费领6万重疾保额，是短期的，如果只靠这个，没有其他商业险，万一真发生重疾，6万能干嘛，解决不了问题，以后的生活开始裸奔。
Reference	(This is yet another trap for the policyholder) It is a short-term activity to claim 60000 serious disease insurance for free; if you only rely on this and there is no other commercial insurance, 60000 yuan will not solve any problem in case of the happening of a serious disease, and you will have nothing in your life later.
Translation	(This is another pit for insurance applicants.) It is a short-term free sum insured for critical illnesses of 60,000 yuan. If you rely only on this and there is no other commercial insurance, what can you do if you really have a critical illness? If the problem cannot be solved, your future life will start running naked.
Remedy-R	Let's evaluate the translation based on the given criteria: 1. Accuracy: – The translation attempts to convey the meaning of the source text, but there are some inaccuracies and awkward phrasing. – "坑" is translated as "pit," which is a colloquial term that might not be as clear in English. "Trap" or "trick" might be more appropriate. – "裸奔" is translated as "running naked," which is a literal translation but doesn't capture the idiomatic meaning. "Struggle to survive" or "barely get by" might be more natural. 2. Fluency: – The translation is mostly understandable but has some awkward phrasing. – "sum insured" is a technical term that might be confusing for a general audience. "coverage" might be more accessible. – "If the problem cannot be solved" is awkwardly phrased. "This won't solve the problem" would be more natural. – "start running naked" is a literal translation but doesn't flow well in English. "barely get by" or "struggle to survive" would be more natural. 3. Completeness: – The translation attempts to convey all the information from the source, but the phrasing is awkward and the idiomatic expressions are not well translated. 4. Language Correctness: – The translation is mostly correct but has some awkward phrasing and idiomatic issues. – "sum insured" is a technical term that might be confusing for a general audience. "coverage" might be more accessible. – "running naked" is a literal translation but doesn't flow well in English. "barely get by" or "struggle to survive" would be more natural. Overall, the translation is mostly understandable but has some issues with fluency and idiomatic expression.
EviSpan	[Annotated Translation] (This is another pit for insurance applicants.) It is a short-term free sum insured for critical illnesses of 60,000 yuan. If you rely only on this and there is no other commercial insurance, what can you do if you really have a critical illness? <v0>If the problem cannot be solved, your future life will start running naked.</v0>. [Errors] v0 Accuracy/Mistranslation Major [Rationale] The phrase "If the problem cannot be solved, your future life will start running naked" is a mistranslation of the original Chinese text. The original text uses a metaphorical expression "以后的生活开始裸奔" which means "your future life will start without any protection," referring to being unprotected or vulnerable. The translation incorrectly renders this as "running naked," which is not idiomatic and does not convey the intended meaning.

Table 21: Qualitative example where Remedy-R repeatedly discusses the literal rendering of an idiomatic expression across several criteria. EviSpan marks the clause containing a literal and non-idiomatic rendering of the Chinese metaphor.